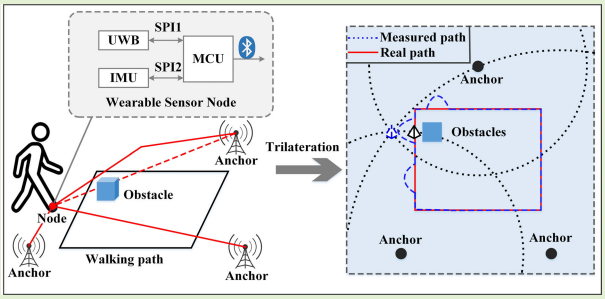
Compensating the NLoS Occlusion Errors of UWB for Pedestrian Localization With MIMU

Zhengyang Qin, Zhaozong Meng, *Member, IEEE*, Zhen Li, *Member, IEEE*, Nan Gao, Zonghua Zhang, Qingyi Meng, and Dong Zhen

***Abstract*—The ultrawideband (UWB) technology has become a competitive choice for indoor pedestrian localization due to its advantages in accuracy, compact size, reliability, power efficiency, and easy setup. However, the none-line-of-sight (NLoS) occlusion caused by the human body parts and obstacles in its propagation path may introduce significant location errors that limit its performance and field of application. To handle the NLoS occlusion errors and promote the localization accuracy for complex indoor environments, a micro-electromechanical system (MEMS) inertial measurement unit (MIMU)-assisted UWB localization system with NLoS occlusion error compensation techniques is provided. First, an NLoS occlusion detection method using the built-in channel information of the UWB transceivers and the variation pattern of anchor-to-node distance is proposed. Second, an extended Kalman filter (EKF)-based attitude computation and a zero-velocity update (ZUPT)-based continuous gait segmentation are introduced for inertial navigation during occlusion intervals. Then, the UWB and IMU parameters are integrated with an unscented Kalman filter (UKF) framework to compensate for the occlusion error and improve the accuracy and robustness. Finally, a battery-powered miniature wearable device is developed and the proposed techniques are verified with experimental studies. The results demonstrate the feasibility and effectiveness of the presented techniques, which provides a good reference for related Internet of Things (IoT) applications.**



***Index Terms*— Extended Kalman filter (EKF), indoor localization, none-line-of-sight (NLoS) occlusion detection, unscented Kalman filter (UKF), zero-velocity update (ZUPT).**

I. Introduction

With the proliferation of the Internet of Things (IoT) techniques in different fields, indoor location information has been an indispensable part of many smart applications [1], [2], including disaster relief, medical treatment, elderly care, and somatosensory gaming. Much effort has been devoted to wireless communication infrastructure-based indoor localization, including Wi-Fi, Bluetooth, Zigbee, and 4G/5G [3]. Different from the other alternatives, ultrawideband (UWB) achieves localization with a subnanosecond ultranarrow pulse, which outperforms its peers in accuracy, compact size, reliability, power efficiency, and easy setup [4], [5]. However, UWB localization relies on the radio wave ranging and trilateration for localization, where none-line-of-sight (NLoS) occlusion in the propagation path of the radio waves may attenuate the signal and the reflected signals may be mistaken for the effective ones for localization. It is a promising way to integrate a miniature micro-electromechanical system (MEMS) inertial measurement unit (MIMU) into a UWB system to compensate for the NLoS occlusion errors [6]. Since the MIMU is competitive in miniature size, simplified system, and anchor-less, its combination with UWB may potentially allow the two techniques to overcome the inherent weakness for each other, namely, the NLoS occlusion error of UWB localization and cumulative error of inertial navigation [7], [8].

Although UWB-IMU combined localization has been an attractive topic for years, the localization techniques for complex indoor environments still deserve further investigation. The key technical challenges facing the related studies include: 1) the identification of UWB NLoS occlusions with

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the limited information that the UWB transceivers provide; 2) the compensation of UWB NLoS occlusion errors using inertial devices that could effectively suppress the cumulative errors during pedestrians' continuous moving; and 3) the effective algorithm framework to integrate UWB and IMU data with acceptable accuracy and robustness. To deal with the above challenges, this investigation aims to improve the accuracy of UWB localization focusing on the identification of NLoS occlusions and the scheme of error compensation with IMU. The UWB built-in channel information and the variation pattern of anchor-to-node distance are employed to identify the NLoS occlusions, and an unscented Kalman filter (UKF)-based data fusion framework is implemented for the compensation of NLoS errors and improvement of accuracy. The main contributions of this investigation include: 1) the proposal of an NLoS occlusion detection method using the built-in channel information of the UWB transceivers and the variation pattern of anchor-to-node distance; 2) the introduction of extended Kalman filter (EKF)-based attitude computation and a zero-velocity update (ZUPT)-based continuous gait segmentation to suppress the cumulative errors of inertial measurement; and 3) the optimization of a UKF-based framework for UWB and IMU data fusion that includes NLoS occlusion variables to accommodate the nonlinear characteristics of pedestrian walking and compensate for the localization errors.

The remainder of this article is structured as follows. Section II surveys the related studies and identifies the key technical challenges. Section III proposes the NLoS occlusion identification and compensation methods. Section IV gives the system implementation and experimental verifications. Finally, Section V concludes the presented investigations with an outlook for future work.

II. Technics and Methods for UWB Indoor Localization

Although UWB has been considered a competitive alternative for indoor applications, there are still technical challenges concerning its accuracy and usability. Much effort has been devoted to handling the intrinsic multipath problems of radio waves, which may introduce serious errors to UWB localization systems, especially in an indoor environment with many obstacles. This section summarizes the related studies and identifies the key technical challenges facing this area.

A. UWB Localization Techniques

The impact of NLoS occlusions in UWB localization can be described with the diagrams in Fig. 1. The UWB system measures the distances d_1, d_2, and d_3 between the moving node N_0 and the three anchors A_1, A_2, and A_3, and computes the location of the node using trilateration. When there is an occlusion between N_0 and A_1, the power of radio waves is attenuated and the wave reflection path A_1-R-N_0 is mistaken for a line-of-sight (LoS) path and the distance between A_1 and A_0 increases d_1 from to $d_{11} + d_{12}$. As a result, an evident error is introduced to the trilateration computing, and the location of N_0 is mistaken for N_0'. To handle the influence of NLoS occlusions, investigations of NLoS occlusion identification and error compensation are conducted and reported in the literature.

1) NLoS Occlusion Identification Techniques: The NLoS occlusion identification techniques can be classified into three categories: distance-based, location-based, and channel information-based. The distance-based method classifies the UWB signal into NLoS or LoS using the variance or the probability density function of the measured distance based on existing measurement data as prior information [9]. The location-based method achieves the localization with a dataset of the anchor-to-node information of known locations, and the result is obtained by comparing the real-time distances with prior locations [10]. The channel information-based method employs the channel impulse response (CIR) with threshold, probability, or machine learning techniques for classification and NLoS occlusion identification [11]. This method is competitive in reliability and accuracy, which has attracted more research effort. Gururaj et al. [12] proposed to identify the NLoS occlusion through the difference of the received UWB power between the practical propagation path and the straight line one. Although it is not competitive in accuracy, real-time localization can be achieved without prior information. Savic et al. [13] supposed that the difference in the received UWB power between the practical propagation path and the straight line conformed to the Gaussian distribution and detected the NLoS occlusion using the probability density distribution function. Ferreira et al. [14] employed the built-in channel information as the feature data for the training and testing of machine learning techniques, and the comparison of different machine learning methods was conducted to obtain the optimal algorithm and feature sets. Jiang et al. [15] integrated convolutional neural network (CNN) and long short-term memory (LSTM) for NLoS/LoS classification with UWB CIR information, and the two methods are used for feature extraction and classification.

From the above discussion, most of the NLoS identification techniques rely on prior information. The use of a large amount of data for feature extraction may make the algorithm time-consuming, which also jeopardizes the adaptivity of the algorithms. It is found that the power of the UWB transceiver received signal is widely used for NLoS occlusion identification, and the other UWB channel information and auxiliary sensors may be promising for occlusion identification with proper feature exploration and decision-making methods.

2) NLoS Occlusion Error Compensation Methods: The compensation of NLoS occlusion error after its identification is also a critical task. Normally, the compensation is implemented through either an NLoS occlusion error compensation model or multisensor data fusion techniques. It is common to use machine learning algorithms to build a relation model for NLoS occlusion error compensation. Otim et al. [16] built a model between the measured distance and human body-node relative location and compensated the NLoS occlusion errors introduced by human body parts with the established model. Dong [17] detected the NLoS occlusions between anchors and a moving node using the Fresnel region between two anchors and alleviated the NLoS error by subtracting the errors relative

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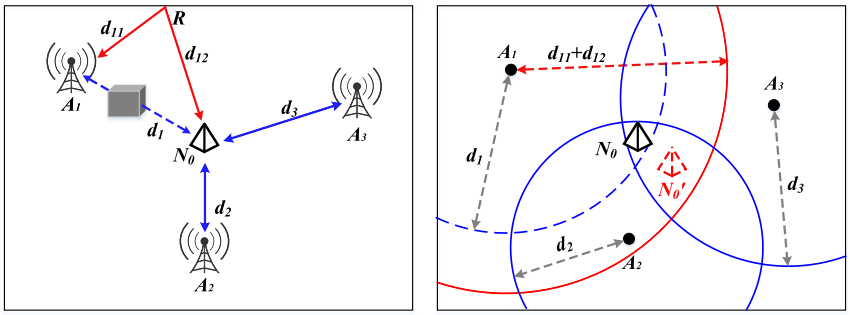


Fig. 1. NLoS occlusion and its influence in UWB localization. (a) UWB localization with NLoS occlusion. (b) Location computation with trilateration.

to LoS, which could reduce 96.21% of the localization errors for a dynamic NLoS environment. Yu et al. [11] introduced the fuzzy comprehensive evaluation (FCE) for NLoS identification using channel information, applied an equality constrained Taylor series robust least squares (ECTSRLS) technique to suppress the NLoS range errors, and demonstrated the performance for harsh indoor environments.

Some customized models for NLoS occlusion identification may not be adaptive to the different application scenarios. A comprehensive framework integrating the NLoS occlusion identification, error compensation, and data fusion of UWB with suitable assistive sensing devices is demanded.

B. UWB-IMU Combined Localization

The integration of UWB and various sensing elements has been considered a promising way to suppress NLoS occlusion errors. With the rapid progress of MEMS and their application in different fields, the MIMU has become a competitive choice for its low profile, power efficiency, independency, and multivariable sensing capability. The data fusion techniques to integrate the outputs of multiple sensors are critical for enhancing the system performance. For instance, Yao et al. [18] improved the performance of the underwater navigation system with a hybrid interacting multiple model (HIMM) to integrate an inertial navigation system (INS) and a Doppler velocity log (DVL). Shen et al. [19] presented a self-learning square-root-cubature Kalman filter (SL-SRCKF) algorithm to improve the seamless navigation of a global positioning system (GPS)/INS system in GPS-denied environments, which achieved a continuous output of accurate results in the absence of partial data. Many novel methods are introduced to UWB-IMU data fusion, including weighted fusion algorithm, figure optimization integrated navigation, Kalman filter, and unbiased finite-impulse response (UFIR).

The weighted fusion algorithm is easy to use, while it requires a high weight factor to generate ideal results. Yang et al. [20] defined the environment factor as the weighted coefficient for the evaluation of the NLoS occlusion in indoor localization. The environment factor is used to merge the location results of inertial navigation estimation with UWB measurement, which eventually promoted the accuracy of NLoS areas by 80% compared to UWB-only systems. Due to the advantages of smooth growth and global optimization, the graph optimization model is increasingly used for multisensor data fusion. Zheng et al. [21] proposed a new UWB-IMU tightly coupled data fusion algorithm based on a graph optimization model to improve the accuracy of system position and heading estimation, which obtained an average attitude error of 2.3° and a root-mean-square error (RMSE) of the positioning result of 4.0 cm. Since the UFIR algorithm is insensitive to noise, UFIR-based UWB-IMU data fusion systems are usually competitive in robustness [22]. Xu et al. [23] proposed a fixed-lag extended finite-impulse response smoother (FEFIRS) for UWB-IMU data fusion, which improved the robustness of the UWB-IMU combined localization system by avoiding the mismatch of the noise model with the practical system. Xu et al. [24] presented a UFIR framework for UWB-IMU data fusion to suppress the influence of UWB colored measurement noise (CMN), which outperformed other solutions in sensitivity and robustness for quadrotor localization under industrial conditions. The Kalman filter algorithm obtains the optimal estimation value through iterative calculations. Compared with UFIR, it is more competitive in accuracy for continuous motion estimation [25]. Kim and Pyun [26] proposed a Kalman filter framework to fuse UWB and IMU data, which adjusted the Kalman gain according to the NLoS degree to effectively compensate for NLoS errors. Li et al [27] proposed an error state Kalman filter algorithm to fuse UWB and IMU data, which could avoid the problem of covariance singularity caused by the redundancy of state variables and realized complete six-degree-of-freedom state estimation and real-time correction of IMU deviation. Feng et al. [28] presented a UWB-IMU multisensor data fusion system based on UKF, which effectively reduced the complexity of the system and was more suitable for nonlinear systems compared to the Kalman filter.

The integration of UWB and IMU is considered an effective way to improve the accuracy of attitude solution and location computing and then enhance the accuracy and robustness of the system. However, it is still a challenge to compensate for NLoS occlusion errors in a complex indoor environment. The establishment of a UWB-IMU data fusion framework considering the complexity, accuracy, and robustness of the system is a common problem that deserves further investigation.

C. Problem Statement

The NLoS occlusion is a common problem restricting the accuracy of UWB indoor localization, and its identification and

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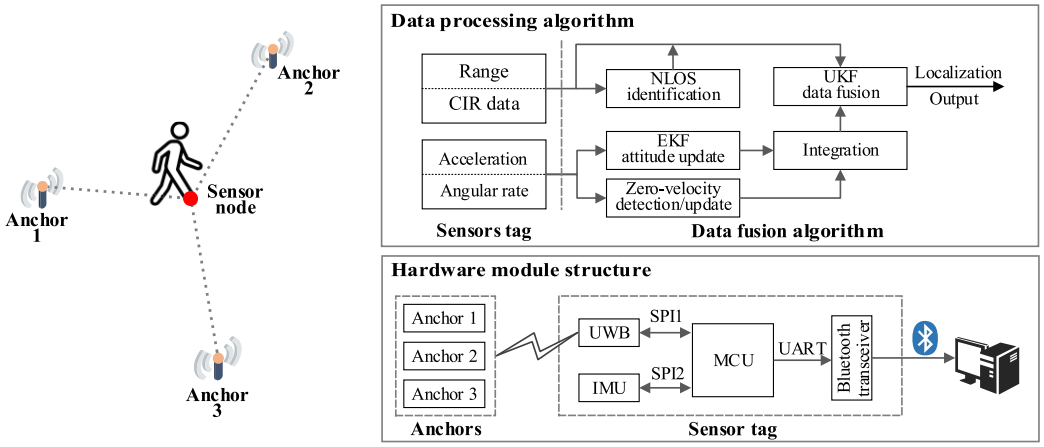


Fig. 2. System scheme for UWB-IMU combined localization.

error compensation techniques have become the focal topics. The key challenges facing this area are summarized as follows.

1) Most NLoS occlusion identification methods rely on prior information that may be hardly adapted to unknown or dynamic environments, which limits the applications of the localization techniques.

2) For UWB-IMU combined localization, the inherent problem of cumulative error for IMU makes it incompetent for the error compensation of long-time occlusion, where an adaptive IMU localization algorithm that is customized to the human body moving is required.

3) Since the UWB-IMU combined localization is affected by uncertainties in a complex localization environment, it is critical to handle the nonlinearity with an integrated framework for occlusion identification, error compensation, and data fusion of UWB-IMU combined localization.

Given the above technical challenges, this investigation aims to provide a data integration framework for IMU-assisted UWB localization with NLoS occlusion identification, error compensation, and UWB-IMU data fusion. First, the UWB built-in channel information and the anchor-to-node distance variation pattern are used to identify NLoS occlusions, and the IMU inertial localization customized to pedestrian walking is applied to compensate for the NLoS occlusion errors. Second, the IMU localization for NLoS occlusion compensation is improved according to the analysis of pedestrian walking patterns to improve the performance. Then, a UKF data fusion framework for UWB-IMU combined localization is constructed to achieve occlusion identification and error compensation and improve the overall accuracy.

III. Identification and Compensation of UWB NLoS Occlusion Errors

A. System Scheme

The principle of the proposed IMU-assisted UWB system regarding its hardware design and data processing can be described with the diagrams in Fig. 2. The sensor node integrates both UWB and IMU with a microcontroller unit (MCU), which maintains a Bluetooth connection with a host PC for visualization. The UWB module collects the ranging of the three anchors and the corresponding built-in channel information for NLoS occlusion identification. Then, the three-axis acceleration and three-axis angular rate are collected and fused with EKF to obtain the attitude and acceleration of the sensor node in the navigation coordinate, and a ZUPT technique is utilized for the segmentation of the pedestrians' continuous walking steps to handle the cumulative error. Finally, the results of the above modules, including occlusion information, ranging, and location, are fused with the UKF to obtain the final location.

B. Detection of UWB NLoS Occlusion

A new NLoS occlusion detection method employing the built-in channel information of UWB transceivers and the variation pattern of anchor-to-node distance is proposed. The CIR information is collected to calculate the received signal power and the first path power, and the difference between the received signal power and the first path power is used as one of the criteria to identify the NLoS occlusions. This method does not need the prior occlusion information of the ambient environment and is also competitive in real-time performance.

Provide that the CIR of anchor i and the node is $R_i = \{C, N, F_1, F_2, F_3\}$, where C is the CIR power, N is the preamble accumulation count value reported by the transceiver, and F_1, F_2, and F_3 are the first path amplitude magnitude values reported by the UWB transceivers. The first path power level denoted by P_{FP} is given by

$$P_{\mathrm{FP}} = 10 \times \log_{10}\left(\frac{F_1^2 + F_2^2 + F_3^2}{N^2}\right) - A. \tag{1}$$

The received power of UWB signal P_{RX} is calculated as

$$P_{\mathrm{RX}} = 10 \times \log_{10}\left(\frac{C \times 2^{17}}{N^2}\right) - A \tag{2}$$

where the constant A is determined by the frequency of the impulse, and the values for 16 and 64 MHz are 113.77 and 121.74, respectively.

Then, define P_D the difference between P_{RX} and P_{FP}, and γ_P the threshold of P_D as criteria 1 to determine the built-in channel information-based NLoS occlusion of the

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UWB transceiver at time k denoted by $N_{k,1}$, where digit 1 means the happening of an occlusion and 0 means no occlusion

$$P_D = P_{\mathrm{FP}} - P_{\mathrm{RX}} \tag{3}$$

$$N_{k,1} = \begin{cases} 1, & P_D < \gamma_P \\ 0, & \text{else.} \end{cases} \tag{4}$$

The variation pattern of the anchor-to-node distance is smooth for LoS UWB localizations, and NLoS occlusions may introduce drastic fluctuations. Therefore, the variation of anchor-to-node distances at time k denoted by s_k is introduced as criteria 2 to determine the anchor-to-node distance-based NLoS occlusions denoted by $N_{k,2}$. Define d as a dataset of the distance between the anchor and the node, and s_k given by (5) is calculated using a k-centered sliding window with $W+1$ elements

$$s_k = \frac{1}{W+1} \sum_{j=k}^{k+W} \left(d_k - \bar{d}\right)^2 \tag{5}$$

$$N_{k,2} = \begin{cases} 1, & s_k < \gamma_D \\ 0, & \text{else} \end{cases} \tag{6}$$

where $\bar{d}$ is the mean of the $W+1$ elements in the sliding window and γ_D is the threshold for s_k.

To improve the accuracy of NLoS occlusion detection, the UWB built-in channel information and anchor-to-node distance variation pattern are integrated. Then, the NLoS occlusion state at time k denoted by N_k is given by the following:

$$N_k = \begin{cases} N_2 \times N_{k-1}, & N_1 + N_2 < 2 \\ 1, & \text{else} \end{cases} \tag{7}$$

where digit 1 means an NLoS occlusion and 0 means no occlusion, and each anchor-to-node channel can be evaluated with N_k to determine the NLoS occlusion state.

C. IMU-Assisted NLoS Error Compensation

To compensate for the UWB localization errors caused by the identified NLoS occlusions with IMU, an algorithm framework shown in Fig. 3 is presented for the data processing of UWB-IMU combined localization. The sensing data of UWB and IMU modules are input to the overall framework integrated with UKF. Before that, the NLoS occlusion identification module determines the NLoS occlusion with UWB data. Meanwhile, an EKF is employed to process the raw IMU data for attitude solution and position computing, and a ZUPT technique is used to handle the inherent cumulative error of inertial navigation for continuous motions.

1) Quaternion-Based EKF for Attitude and Location Computing: The accuracy of attitude is critical for inertial navigation. The quaternion-based EKF is employed to obtain the three-axis attitude, acceleration, and velocity in the navigation coordinate. Let $q_k = [q_0\ q_1\ q_2\ q_3]^T$ be the state variable in quaternion at time k, and the state transition equation is given by

$$q_{k+1} = F_k q_k + w^E \tag{8}$$

where w^E is the noise model conforming Gaussian distribution with zero mean and covariance Q^E and F_k is the state transition matrix at k given by (9). ω_x, ω_y, and ω_z are the angular rate and Δt is the interval between data samples

$$F_k = \frac{1}{2} \begin{bmatrix} 1 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 1 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 1 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 1 \end{bmatrix} \Delta t. \tag{9}$$

Let $a_k^B = [a_x\ a_y\ a_z]^T$ be the observation variable, namely, the three-axis acceleration of the IMU in the carrier coordinate, and v^E is the measurement noise. The measurement equation is given by

$$a_k^B = a_k^{B\prime} + v^E = H^E q_k + v^E \tag{10}$$

where H_k is the Jacobian matrix of the nonlinear measurement function at time k given by (11) and v^E is the zero-mean measurement noise with a covariance R^E

$$H^E = \begin{bmatrix} 2gq_2 & -2gq_3 & 2gq_0 & -2gq_1 \\ -2gq_1 & -2gq_0 & -2gq_3 & -2gq_2 \\ -2gq_0 & 2gq_1 & 2gq_2 & 2gq_3 \end{bmatrix}. \tag{11}$$

According to the principle of EKF, the algorithm flow for attitude estimation is given by Algorithm 1.

With the optimal estimation of q_k and the acceleration in the carrier coordinate a_k^B, the acceleration in the navigation coordinate a_k^N can be obtained with the following:

$$a_k^N = q_k a_k^B q_k^* - \begin{bmatrix} 0 & 0 & g \end{bmatrix}^T. \tag{12}$$

Then, the three-axis velocity v_k and the displacement within Δt in the navigation coordinate denoted by p_k can be given by

$$v_k = v_{k-1} + \frac{1}{2}\left(a_{k-1}^N + a_k^N\right)\Delta t \tag{13}$$

$$p_k = p_{k-1} + \frac{1}{2}\left(v_{k-1} + v_k\right)\Delta t. \tag{14}$$

However, the cumulative error of IMU navigation makes it unacceptable for the continuous motions of pedestrian walking. An effective solution is ZUPT, which identifies the gait cycles, segments the gait cycles into stance and swing phases, and corrects the velocity errors to improve the localization accuracy. A ZUPT technique is employed to reduce the cumulative error.

2) ZUPT for IMU Localization During NLoS Occlusions: The sensing device is usually mounted on the foot of a pedestrian. The human motions in indoor environments are normally walking or running, which are continuously moving where the error of attitude and velocity obtained by the integration of angular rate and acceleration will accumulate with time. The ZUPT technique detects the zero-velocity point and segments the continuous gait cycles into stance and swing phases to correct the cumulative errors. The sensing device collects the acceleration and angular rate of the sensing module mounted on a pedestrian's foot and performs zero-velocity detection with the raw sensor data. The generalized maximum likelihood ratio detection algorithm is used in this investigation to integrate the raw acceleration and angular rate to construct

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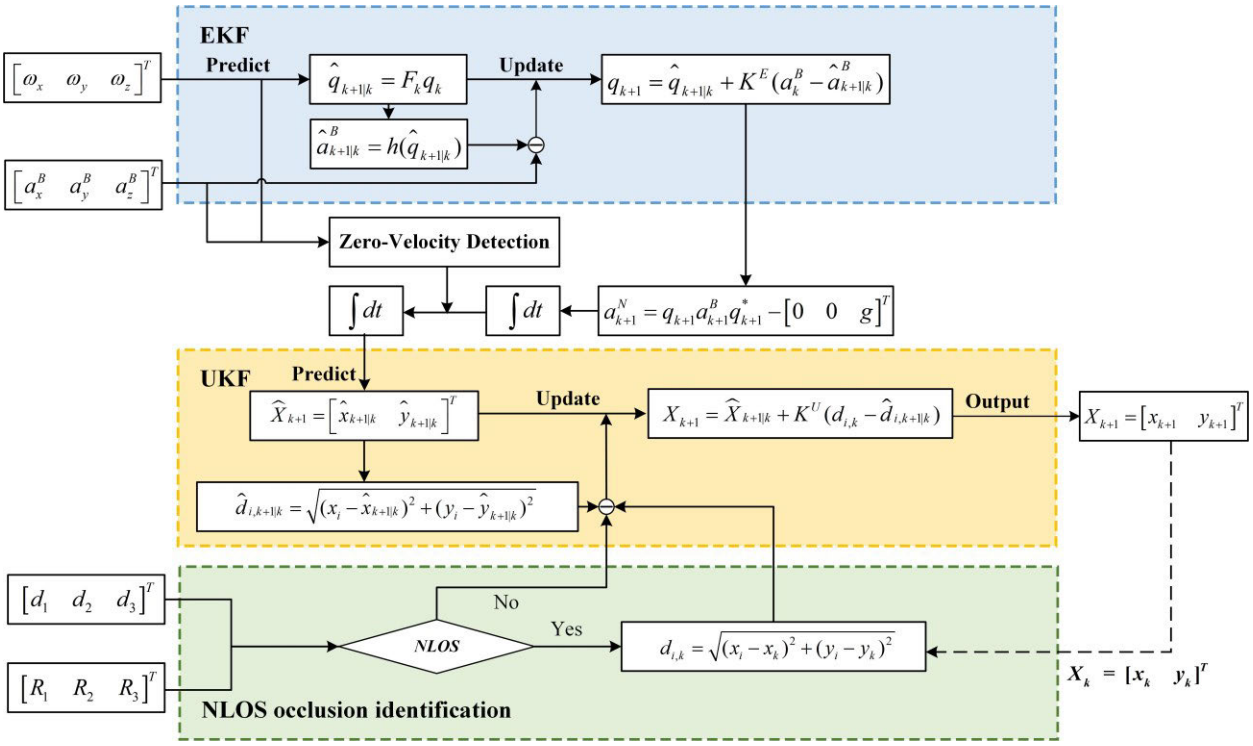


Fig. 3. Algorithm framework for UWB-IMU combined localization.

statistical values to be used for determining the gait cycle phases.

Suppose that $y_k = \{y_{a,k} y_{\omega,k}\}$ are the data samples at time k, which include the three-axis acceleration $y_{a,k}$ and three-axis angular rate $y_{\omega,k}$, and a k-centered dataset with $W+1$ elements is established for calculating the detection statistic T_k with

$$T_k = \frac{1}{W+1} \sum_{i=k-W/2}^{k+W/2} \left(\frac{1}{\sigma_a^2} \left\| y_{a,i} - g \frac{\overline{y_a}}{\|\overline{y_a}\|} \right\|^2 + \frac{1}{\sigma_\omega^2} \|y_{\omega,i}\|^2 \right) \quad (15)$$

where σ_a^2 and σ_ω^2 are the variances of acceleration and angular rate noises and $\bar{y}_a$ is the mean of the accelerations in the sliding window at time k. The detection statistic T_k is compared with a preset threshold β to determine whether the motion state at the current time is in the zero-velocity interval. If the detection statistic T_k is less than the threshold β, it is determined that the current motion state is in the stance phase and the corresponding binary state presented with C_k is given by

$$C_k = \begin{cases} 1, & T_k < \beta \\ 0, & \text{else.} \end{cases} \quad (16)$$

The changes in the pedestrian's gait model during actual movement affect the value of the statistic T_k. A threshold β that is too large will misjudge the swing phase as a stance phase, while a threshold that is too small will result in a false alarm that misjudges the stance phase as a swing phase. Therefore, an adaptive threshold-based ZUPT technique that could dynamically adjust the threshold according to the different gait cycle patterns is demanded. The threshold at time k could be determined by the features of the pedestrian motion modality, including the mean and variance of the detection statistic T_k [29], [30]. An adaptive threshold-based ZUPT technique is introduced to handle the different gait cycle patterns and the dynamic threshold is calculated with the features of the pedestrian motions in the following way.

Algorithm 1 EKF-Based Attitude Estimation

Input	q_k, P_k, a_k, Q^E, R^E
Prediction	Prediction of the state variables and state estimation error for the next iteration.
	$\widehat{q}_{k+1\|k} = F_k q_k$
	$\widehat{P}_{k+1\|k} = F_k P_k F_k^T + Q^E$
Update	Calculation of the Kalman gain.
	$\overset{B}{}_{k+1\|k} = H^E \widehat{q}_{k+1\|k}$
	$K_{k+1}^E = \widehat{P}_{k+1\|k} (H_k^E)^T (H_k^E \widehat{P}_{k+1\|k} (H_k^E)^T + R^E)^{-1}$
	Update of the state variable and state estimation error.
	$q_{k+1} = \widehat{q}_{k+1\|k} + K_k^E (a_k^B - \widehat{a}_{k+1\|k}^B)$
	$P_{k+1} = \widehat{P}_{k+1\|k} - K_k^E H_k^E \widehat{P}_{k+1\|k}$
Return	q_{k+1}, P_{k+1}

Define T as a sliding window of the detection statistics dataset with the center element T_k and a window length $W+1$. The mean and variance of the dataset T are calculated as follows:

$$\mu_k = \frac{1}{W+1} \sum_{i=k-W/2}^{k+W/2} T_i \quad (17)$$

$$\sigma_k^2 = \frac{1}{W+1} \sum_{i=k-W/2}^{k+W/2} (T_i - \mu_k)^2. \quad (18)$$

The adaptive threshold at k denoted by β_k is calculated as

$$\beta_k = \mu_k + K\sigma_k^2 + \alpha \quad (19)$$

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where K and α are an empirical coefficient and constant, respectively, both of which need to be specifically determined to customize the IMU models of different performances.

3) UKF Data Fusion for UWB-IMU Combined Localization: The UKF algorithm is a deterministic sampling-based technique, and its main idea is to describe the Gaussian distribution of variables with a group of sigma sampling points to obtain the approximate probability density distribution of nonlinear functions with the probability density of sampling points. The unscented transform (UT) is a critical step for sigma sampling point calculation. Compared with KF and EKF, UKF is more competitive in handling nonlinearity in the UWB-IMU data fusion. According to the principles of UKF, the algorithm flow of UWB-IMU data fusion is given by Algorithm 2.

The sigma sampling points obtained with UT transform are used to present the features of the original data. The dataset of the sigma points $\{X_i\}$, their weight of mean w_i^m, and the weight of covariance of sigma points w_i^c are given by

$$\begin{cases} X_0 = \overline{x} \\ X_i = \overline{x} + \left(\sqrt{(n+\lambda)P}\right)_i, \quad i = 1, 2, \ldots, n \\ X_{i+n} = \overline{x} - \left(\sqrt{(n+\lambda)P}\right)_i \end{cases} \tag{20}$$

$$w_i^m = \begin{cases} \lambda/(n+\lambda), & i = 0 \\ 1/2(n+\lambda), & i \neq 0 \end{cases} \tag{21}$$

$$w_i^c = \begin{cases} \lambda/(n+\lambda) + 1 + \beta - \alpha^2, & i = 0 \\ 1/2(n+\lambda), & i \neq 0 \end{cases} \tag{22}$$

where $\overline{x}$ is the mean of the sigma points $\{X_i\}$, n is the dimension of the state variable, and P is the covariance matrix. Parameter λ is a scaling parameter to reduce the total prediction error, which is given by

$$\lambda = \alpha^2 (n + \kappa) - n \tag{23}$$

where the constant α is the scale factor to define the range size of the sample point set, which can be set to zero by default, the constant β is used to adjust the approximate accuracy of the covariance with a default value of two for Gaussian distribution, and κ is an empirical constant with a default value of zero.

The UWB-IMU data fusion for the IMU-assisted UWB localization is accomplished with the UKF algorithm. Define $X_k = [x\ y]^T$ the state variable, which denotes the location in the navigation coordinate of the moving sensor node at time k. The state transition equation is given as follows:

$$X_{k+1} = f(X_k) + w^U \tag{24}$$

where $f(\cdot)$ is the nonlinear state transition function, which represents the process of obtaining the location by the IMU with EKF and integration. w^U is the process noise for which the mean is zero and the covariance matrix is Q^U.

The anchor-to-node distances $Z_k = [d_0\ d_1\ d_2]^T$ are used as the measurement variable, and the measurement equation is given by

$$Z_k = h(X_k) + v^U \tag{25}$$

where the nonlinear functions in measurement equation $h(\cdot)$ are given by (26) and v^U is the zero-mean measurement noise with covariance denoted by R^U

$$\begin{cases} d_0 = \sqrt{(x_0 - x)^2 + (y_0 - y)^2} \\ d_1 = \sqrt{(x_1 - x)^2 + (y_1 - y)^2} \\ d_2 = \sqrt{(x_2 - x)^2 + (y_2 - y)^2}. \end{cases} \tag{26}$$

In (26), the variables (x_0, y_0), (x_1, y_1), and (x_2, y_2) are the coordinates of the three UWB anchors A_0, A_1, and A_2.

IV. Implementation and Experimental Evaluation

In this section, a wireless sensing module integrating UWB and IMU for pedestrian wearable localization is developed, and a series of tests are conducted to verify the feasibility and performance of the presented techniques, including the NLoS occlusion detection, ZUPT-based gait cycle segmentation, and the performance of the system in handling NLoS occlusions.

Algorithm 2 UKF-Based UWB-IMU Data Fusion

Input	X_k, P_k, Z_k, Q^U, R^U
UT transform	Calculation of Sigma points data set $\{X_i\}$ of the state variable X_k, the weight of mean w_i^m and the weight of covariance of Sigma points w_i^c.
Prediction	Prediction of the state and state estimation error covariance for the next iteration. $\widehat{X}_{k+1\|k} = \sum_{i=0}^{2n} w_i^m f(X_i) P_{k+1\|k} = \sum_{i=0}^{2n} w_i^c [f(X_i) - \widehat{X}_{k+1\|k}][f(X_i) - \widehat{X}_{k+1\|k}]^T + Q^U$
UT transform	Update of Sigma points data set $\{\zeta_i\}$ of the state predictive value $\hat{X}_{k+1\|k}$, the weight of mean $(w_i^m)'$ and the weight of covariance of the Sigma points $(w_i^c)'$.
Update	Estimation of the covariance of the predicted measurement $P_{zz,k+1\|k}$ and the cross-covariance between $\hat{X}_{k+1\|k}$ and $\hat{Z}_{k+1\|k}$ denoted by $P_{xz,k+1\|k}$. $\widehat{Z}_{k+1\|k} = \sum_{i=0}^{2n} w_i^{m'} h(\zeta_i) P_{zz,k+1\|k} = \sum_{i-0}^{2n} w_i^{c'} [h(\zeta_i) - \widehat{Z}_{k+1\|k}][h(\zeta_i) - \widehat{Z}_{k+1\|k}]^T + R^U$ $P_{xz,k+1\|k} = \sum_{i=0}^{2n} w_i^{c'} [f(X_i) - \widehat{X}_{k+1\|k}][h(\zeta_i) - \widehat{Z}_{k+1\|k}]^T$ Calculation of the Kalman gain. $K_{k+1} = P_{xz,k+1\|k}(P_{zz,k+1\|k})^{-1}$ Update of the state and state estimation error covariance. $X_{k+1} = \widehat{X}_{k+1\|k} + K_{k+1}(Z_k - \widehat{Z}_{k+1\|k}) P_{k+1} = P_{k+1\|k} - K_{k+1} P_{zz,k+1\|k} K_{k+1}$
Return	X_{k+1}, P_{k+1}

A. UWB-IMU Wireless Sensor Node

For the wearability purpose, the hardware module to be developed is expected to be low-profile, power-efficient, and wireless data transfer. The hardware module consisting of an MCU, an IMU, a UWB transceiver, and a Bluetooth transceiver is designed and assembled, as shown in Fig. 4.

The key parameters of the selected components are given in Table I. Considering the key requirements, the selected MCU

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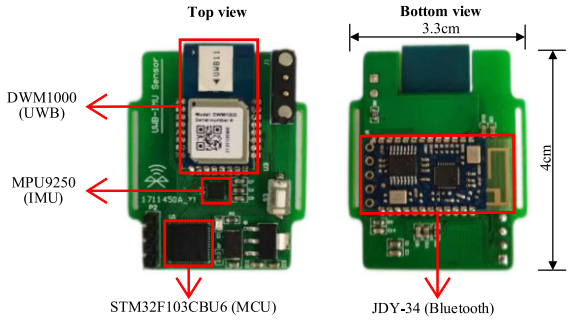


Fig. 4. Hardware module developed.

TABLE I
SPECIFICATIONS OF THE KEY COMPONENTS

Components	Model	Power	Key parameters
MCU	STM32F103 CBU6 (STMicro., EU)	2.0~3.6V	ARM 32-bit Cortex™-M3 CPU CPU frequency 72MHz QFN footprint (7×7mm)
IMU	MPU-9250 (InvenSense, US)	2.4~3.6V	Accel. full scale range: ±4g Accel. sensitivity: 1/4096g/LBS Accel. rate noise: 300μg/√Hz Gyro full scale range: ±1000°/s Gyro sensitivity: 1/32.8°/s/LBS Gyro rate noise: 0.01°/s/√Hz SPI interface (20MHz max)
UWB transceiver	DWM1000 (DecaWave, Ireland)	2.8~3.6V	IEEE802.15.4-2011 UWB Asset location to a precision of 10cm
Bluetooth transceiver	JDY-34 (Risym, China)	2.1~3.6V	Transmission rate 40K Byte/s Baud rate 115200bps

is STM32F103CBU6 by STMicroelectronics, EU; the selected IMU is MPU-9250 by InvenSense, USA; and the selected UWB transceiver is DWM1000 by DecaWave, Ireland. The DWM1000 module integrates the DW1000 UWB IC, antenna, clock control, and power management. Then, the selected Bluetooth transceiver is JDY-34 by Risym, China.

For this IMU-assisted UWB localization system, the MCU collects the UWB and IMU data with a sampling rate of 100 Hz. The raw data are collected and sent to the host PC for the rest processing with Bluetooth at a baud rate of 115 200 b/s. Then, the PC performs data processing, error compensation, displacement calculation, and the display of motion trajectory.

B. NLoS Occlusion Identification

Test 1 verifies the feasibility of the NLoS error identification technique, which employs the channel information of the UWB transceiver and the variation pattern of the anchor-to-node distance to identify the NLoS occlusions. The experimental setup is shown in Fig. 5. A sensing node is installed on a slide rail platform, and an obstacle is placed between the signal transmission path between the anchors and the sensing node. When the slide rail platform drives the node to move from right to left, the built-in channel information and the distances are collected for the analysis of NLoS occlusions.

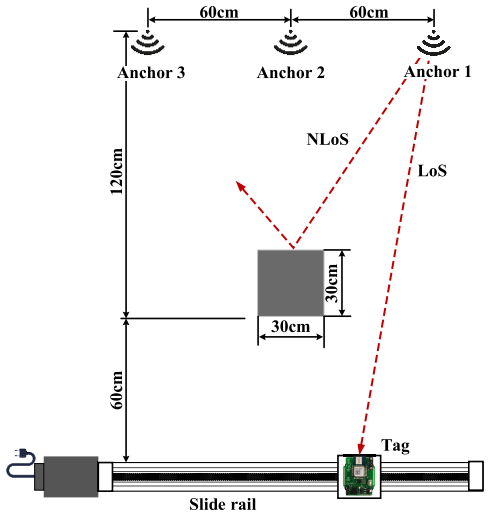


Fig. 5. Setup for Test 1.

With the Bluetooth transceiver of the sensing node, the UWB channel information and ranging information are collected by the MCU to determine the NLoS occlusions. The received power P_{RX}, the first path signal power P_{FP}, and the power difference P_D can be obtained with (1)–(3). As shown in Fig. 6(a), the red, orange, and blue lines denote the power difference P_D of the UWB signals from anchors 1, 2, and 3, respectively. As shown in Fig. 6(b), the red, orange, and blue curves represent the distance of sensor node to anchors 1, 2, and 3, respectively. The black line in binary logic represents the NLoS occlusion determined by the proposed method, where the logic high indicates that it is in NLoS occlusion, and the logic low indicates that there is no NLoS occlusion. When the slide rail platform drives the sensor node from right to left, the obstacle introduces NLoS occlusion in the actual measurement. It is clear that the values of P_D and d change drastically when the UWB signal is obscured by obstacles and they will become smooth when it returns to LoS transmission.

For the test results of NLoS identification, Fig. 6(c), (f), and (i) gives the power difference P_D of UWB signals from anchors 3, 2, and 1, respectively. The NLoS results determined with the power difference P_D only fluctuate drastically and there are evident misjudgments. Fig. 6(d), (g), and (j) gives the variations of anchor-to-node distances between the node and anchors 3, 2, and 1, respectively. The NLoS results determined with the anchor-to-node distance only are stable, but there are also misjudgments intervals. The determined NLoS error time intervals overtake the actual occlusion time intervals. Therefore, the final decision integrates the two results with (7). Then, Fig. 6(e), (h), and (k) gives the final NLoS decisions of the occlusion states of anchors 3, 2, and 1, respectively. The results show that the UWB channel information combined with the anchor-to-node distance variation pattern can improve the accuracy and stability of NLoS occlusion identification.

C. ZUPT-Based Gait Cycle Segmentation

Test 2 verifies the performance of the ZUPT algorithm in segmenting the gait cycle phases of pedestrian walking. In this test, the wireless sensor node is mounted on one of

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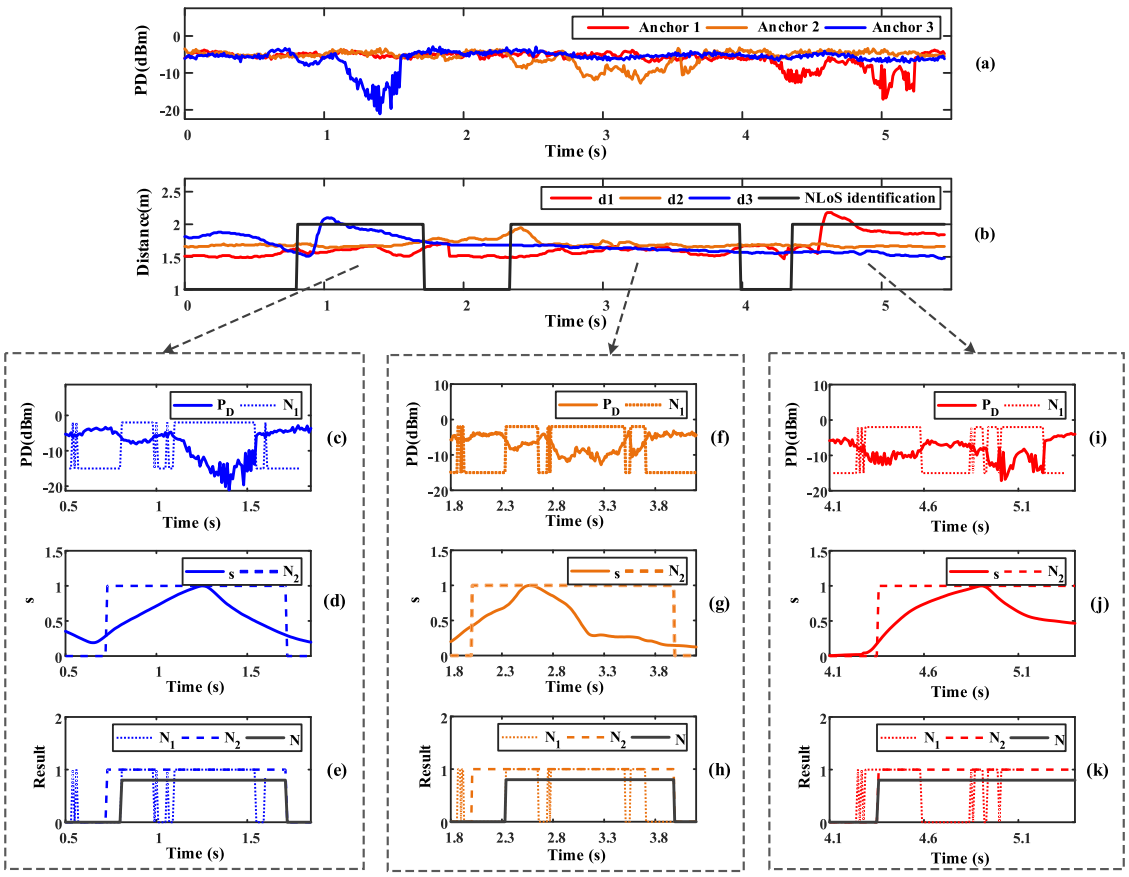


Fig. 6. Decision-making of NLoS occlusion identification. (a) P_D of the UWB signals from anchors. (b) anchors to node distance and the NLoS occlusion identification results. (c)–(e) P_D, anchor-node distance, and NLoS occlusion decision of anchor 3. (f)–(h) P_D, anchor-node distance, and NLoS occlusion decision of anchor 2. (i)–(k) P_D, anchor-node distance, and NLoS occlusion decision of anchor 1.

the pedestrian's shoes, and the three-axis acceleration and three-axis angular rate of the IMU are collected at a frequency of 100 Hz during the pedestrian's walking. The statistical value T is calculated with (15), and the value of T is then compared with the adaptive threshold to segment the pedestrian's walking gait cycles into stance and swing phases. The result of the ZUPT-based gait cycle segmentation is shown in Fig. 7. Fig. 7(a) and (b) shows the raw data of the three-axis acceleration and angular rate acquired during pedestrian walking. Then, Fig. 7(c) shows the statistic T calculated by the generalized maximum likelihood ratio algorithm using the raw data of acceleration and angular rate, as well as the gait cycle segmentation results. The red dashed line represents the statistics by the generalized maximum likelihood ratio detection algorithm, and the black line in binary logic represents the determined gait cycle phases. Logic 1 stands for the stance phases and logic 0 stands for the swinging phases.

To verify the adaptivity of the ZUPT algorithm, experiments were designed to evaluate the performance of the algorithm for different walking speeds. A pedestrian is requested to move along the same path with the same step length at three speeds: quick, medium, and slow. The speeds for quick, medium, and slow are about 2.0, 1.5, and 1.2 s/step. Fig. 8 shows the zero-velocity detection results of the generalized maximum likelihood ratio-based detection algorithm for the three different motion speeds. In the figure, the red line represents the detection statistics, the blue line represents the adaptive threshold, and the black line represents the determined gait cycle phases. The experiment results show that for different walking speeds, the number of gait cycles detected by the zero-velocity detection algorithm agrees well with the actual number of walking steps. The accuracy and stability of the presented zero-velocity detection algorithm are acceptable for practical pedestrian indoor localization applications.

To evaluate the performance of the ZUPT algorithm, the detection result of the generalized maximum likelihood ratio-based detection algorithm is compared with the square sum of the accelerations-based detection algorithm. Choosing one segment of the sensing data as an example, the gait cycle segmentation results of the above two methods are shown in Fig. 9. As shown in Fig. 9(a), the red line represents the statistic T that is obtained with the generalized maximum likelihood ratio-based algorithm, and the black line in binary logic represents the determined gait cycle phases. Logic 1 represents the stance phase and logic 0 represents the swing phase. In contrast, the blue line in Fig. 9(b) represents the statistic T that is obtained with the square sum-based detection algorithm, and the black line is the determined gait cycle phases in binary logic as well. The results show that the gait cycle phases determined by the generalized maximum likelihood ratio-based algorithm are in good agreement with the actual number of walking steps. Since the square sum-based

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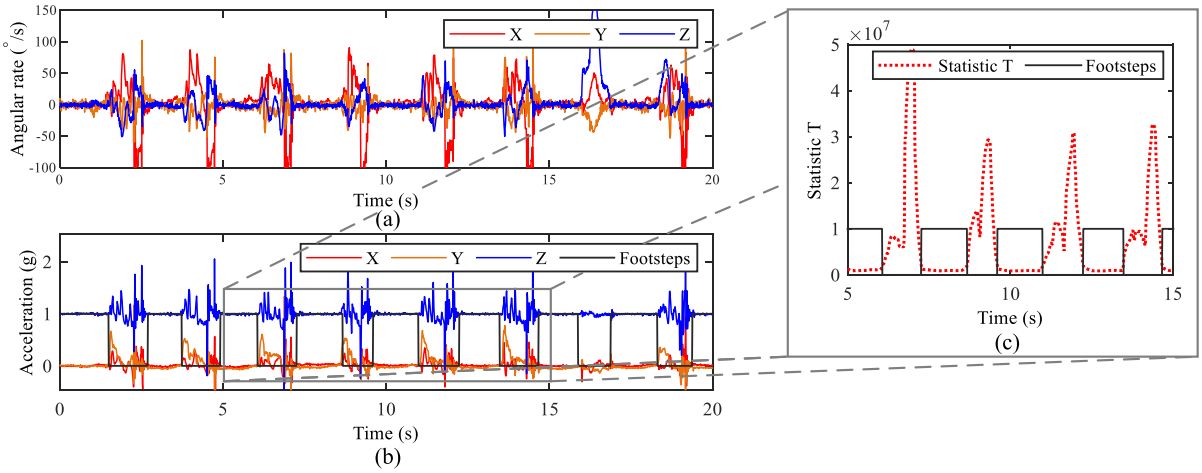


Fig. 7. Results of gait cycle segmentation with ZUPT. (a) Raw data of tri-axial angular rate. (b) Raw data of tri-axial acceleration and the results of gait cycle segmentation. (c) Generalized maximum likelihood ratio detection statistics and the gait segmentation results.

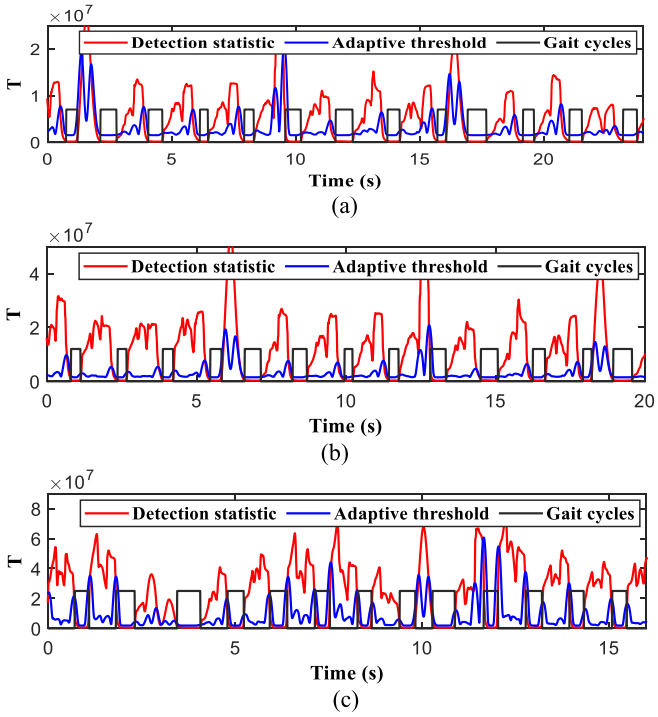


Fig. 8. Zero-velocity detection for three different walking speeds. (a) Slow. (b) Medium. (c) Quick.

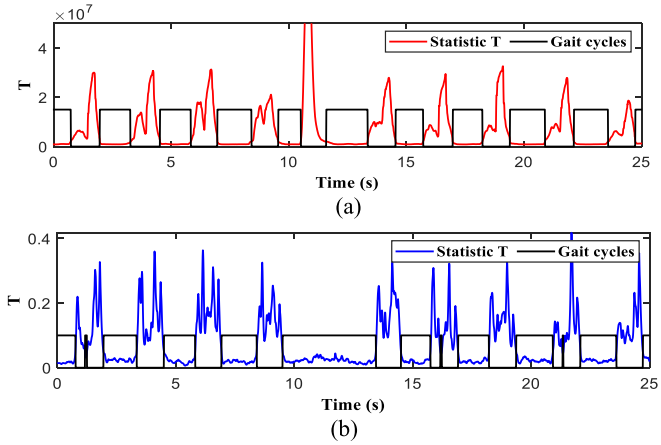


Fig. 9. Comparison of zero-velocity detection methods. (a) Generalized maximum likelihood ratio-based detection. (b) Acceleration square sum-based detection.



Fig. 10. Test scene.

algorithm relies on the feature of the accelerations square sum only, the limited feature may lead to misjudgment. Therefore, the accuracy and stability of the former outperform the latter in zero-velocity detection for gait cycle segmentation purposes.

D. Evaluation of the Performance With NLoS Occlusions

Test 3 verifies the performance of the IMU-assisted UWB localization for an indoor localization environment with NLoS occlusions. The test scene with concrete pillars as obstacles and the arrangement of the anchors are shown in Fig. 10. The UWB-IMU wireless sensor node is mounted on the pedestrian's left shoe, and the pedestrian is requested to walk following a predefined path. In addition to the concrete pillars, the pedestrian's lower limbs are also obstacles to introducing NLoS occlusions for the UWB signal transmission.

First, the evaluation of the impact of NLoS occlusions on the accuracy of UWB positioning is conducted with tests of both an LoS path and a path with NLoS occlusions, and the results are shown in Fig. 11(a) and (b). The black line is the predefined reference path and the blue line is the measured trajectory. It is easy to find that there are

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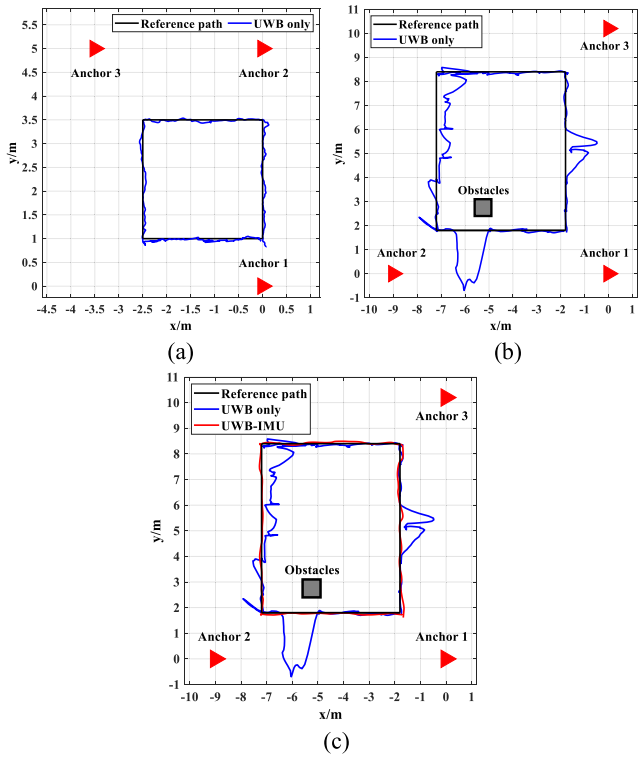


Fig. 11. Performance of the UWB-IMU combined localization. (a) LoS path. (b) Path with NLoS occlusions. (c) NLoS path after error compensation.

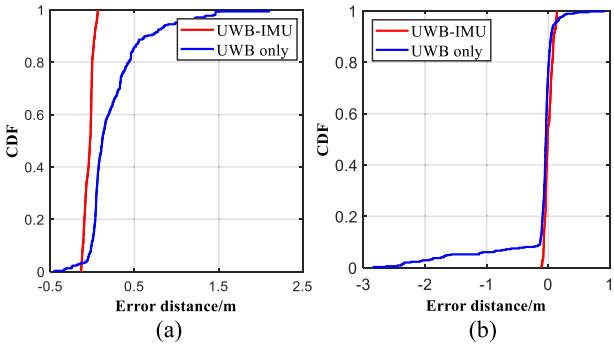


Fig. 12. CDF of position error. CDF of (a) x-axis and (b) y-axis.

TABLE II
TEST RESULTS OF THE SPIRAL PATH EXPERIMENT

	X-axis		Y-axis	
	Average (cm)	RMSE (cm)	Average (cm)	RMSE (cm)
1	6.60	10.92	6.70	17.28
2	7.43	5.89	7.02	10.16
3	6.17	7.38	7.94	6.62
4	5.02	11.37	6.57	7.48
5	5.35	7.10	7.86	14.30

significant errors in UWB localization with NLoS occlusions, which makes it unacceptable for practical use. The key factors recognized to affect the localization accuracy include the alternating footsteps that make the lower limbs temporarily block the UWB signals, obstacles such as concrete pillars to block UWB signals, UWB device accuracy, and the deviation between the actual motion path and the predefined path.

Second, the experimental evaluation of the NLoS occlusion error compensation techniques is conducted as well. The pedestrian wears the UWB-IMU wireless sensor node on the left foot and is requested to walk counterclockwise along the predefined path shown in Fig. 11(c). The black line is the predefined reference path, and the red and blue lines in the figure are the measured trajectories with and without NLoS error compensation. From the results, the positioning of UWB suffers from the NLoS occlusions that block the UWB signal transmission, which introduces significant deviation from the predefined path, and the NLoS occlusion identification and error compensation techniques can effectively suppress the NLoS occlusion errors.

The cumulative distribution functions (CDFs) of the location errors for x-axis and y-axis are given in Fig. 12. Specifically, the maximum location error in the x-axis and y-axis of the UWB localization system without NLoS compensation is over 2.00 m. The average position errors of the x-axis and y-axis are about 27.00 and 19.02 cm, respectively. For the proposed NLoS occlusion identification and error compensation techniques, the maximum position error of the x-axis and y-axis is about only 13.00 cm. The average position errors of the x-axis and y-axis are about 5.07 and 5.33 cm, respectively. The localization accuracy in both x-axis and y-axis has been reduced by over 70%.

Then, the evaluation of the presented techniques and system in the case of long-distance continuous pedestrian walking is evaluated through experiments with a spiral path. The pedestrian is requested to perform five repeated walking tests and the data are collected for further analysis. Fig. 13 shows the predefined spiral path as the reference and the measured trajectories of the five repeated tests as the localization results.

After analyzing the positioning accuracy of the five localization results, the average error and RMSE of the x-axis and y-axis are shown in Table II. In this experiment, the proposed techniques can effectively identify and compensate for NLoS errors, and the error does not increase over time. The average errors of the system in the x-axis and y-axis are 6.11 and 7.22 cm, respectively, and the RMSEs of the x-axis and y-axis are 8.53 and 11.17 cm, respectively. The proposed NLoS occlusion identification and error compensation techniques can effectively suppress the NLoS occlusion errors and achieve acceptable localization accuracy for indoor environments with NLoS occlusions. To evaluate the performance of the algorithm for practical applications implemented with a resource-limited MCU, the experimental verification of the algorithm complexity of the proposed presented system is also conducted. By a test using the cycle count of the MCU development kit, the execution time of the algorithms for the functions of NLoS detection, EKF attitude computation, and UKF data fusion is about 4.60 ms under a system clock 72 MHz. The time complexity of the algorithm satisfies the requirement for real-time applications.

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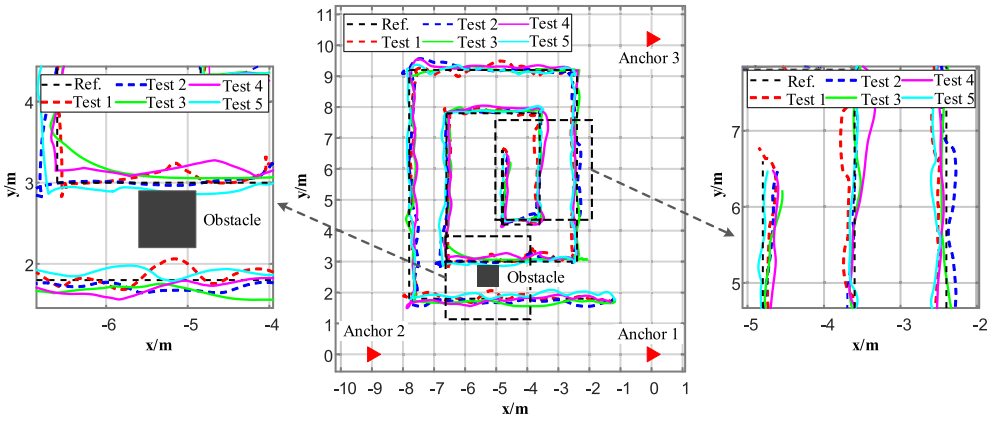


Fig. 13. Experimental evaluation with a spiral path.

V. Conclusion

In this investigation, an IMU-assisted UWB localization system with NLoS occlusion identification and error compensation techniques is proposed. The NLoS occlusions of UWB devices are identified with the channel information and anchor-to-node distance variation pattern. The localization functions are implemented by fusing the UWB and IMU data with a UKF framework. The system adjusts the data fusion according to the extent to which the UWB signal is affected by NLoS error at the moment. Meanwhile, a zero-velocity detection technique is employed to improve the accuracy of inertial measurement by segmenting the pedestrian gait cycles into stance and swing phases and correcting the velocity errors accordingly. A series of experimental studies are conducted and the test results prove that the system can effectively identify and compensate for NLoS errors of UWB devices. The experiment also demonstrates the adaptivity and repeatability of the presented techniques for complex indoor environments. The experiment results also prove that the generalized maximum likelihood ratio-based zero-velocity detection is effective to suppress the intrinsic cumulative errors of inertial measurement, which is helpful to improve the overall accuracy. Since the sensor device is small and wireless, it could implement the functions without interfering user's daily activities. The presented work has provided a good reference for nonintrusive wearable sensing and related IoT applications.

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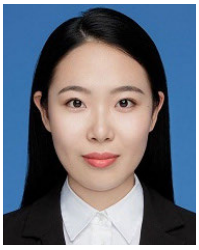
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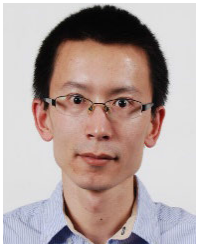
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